Vel Tech
Rangarajan Dr. Sagunthala
R&D Institute of Science and Technology
(Deemed to be University Estd. u/s 3 of UGC Act, 1956)

Batch No: MP-AIDS-17

School of Computing

Department of Artificial Intelligence and Data Science

10214AD701- MAJOR PROJECT

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Major Project - Abstract Form

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TITLE & ABSTRACT:

Title: Real-Time Suspicious Activity Detection Using CNN-LSTM ~~and Anomaly Detection~~

Abstract: SafeCrowd Vision is an AI-based real-time surveillance system designed to enhance safety in densely crowded public spaces such as railway stations, markets, religious gatherings, and stadiums. Traditional monitoring methods often depend on human vigilance, which can be unreliable in high-density environments due to fatigue and oversight. To overcome these limitations, this project leverages advanced deep learning techniques to automatically detect suspicious activities and potential theft, significantly improving situational awareness and response capabilities.

The system combines Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) networks to analyze both spatial and temporal aspects of crowd behavior. CNNs extract key visual features from video frames—such as people, objects, and movements—while LSTMs analyze sequences of frames to detect anomalies like sudden running, tailgating, object grabbing, or unauthorized access. To further enhance accuracy, the system uses anomaly detection to compare real-time behaviors with learned normal patterns and integrates object tracking algorithms like YOLO and DeepSORT for identity tracking across frames.

SafeCrowd Vision operates in real time and issues automatic alerts to security personnel via mobile devices or dashboards when threats are detected, ensuring quick and proactive responses. Its flexibility allows integration with fixed CCTVs, drones, or mobile surveillance units. Trained on diverse datasets containing real and synthetic suspicious activities, the system is robust and adaptable to various environments, offering a scalable and intelligent solution to modern crowd safety challenges.

Keywords: Suspicious Activity Detection, Crowd Surveillance, Deep Learning, Convolutional Neural Networks (CNN), Long Short-Term Memory (LSTM), Anomaly Detection, Computer Vision, Theft Detection, Real-Time Monitoring, Smart Surveillance, Behavior Analysis.

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